

2026 Testing Policy

Uni Textbook Marketplace

For **Project Day**

Presented by:

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Testing Policy

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Contents

1 Introduction **3**

2 Testing Objectives **4**

3 Testing Types **5**

3.1 Unit Testing 5

3.2 Integration Testing 5

3.3 End-to-End (E2E) Testing 6

3.4 Manual and Exploratory Testing 6

3.5 Accessibility Testing 6

4 Testing Tools and Environments **8**

4.1 Environments 8

5 Coverage Targets **10**

6 Defect Management Process **11**

6.1 Workflow 11

6.2 Severity Levels 11

7 Acceptance Criteria **13**

1. Introduction

This document lays out how we approach testing on the Uni Textbook Marketplace project: what we test, how we test it, what tools we use, how we handle bugs when we find them, and what has to be true before a feature is considered done. The point of writing this down isn't to look impressive, it's so all four of us are actually testing things the same way instead of everyone deciding their own bar for "good enough" on their own features.

Like the Coding Standards document, most of what's in here isn't just policy on paper, it's enforced by our CI pipeline. If a pull request doesn't have passing tests, it doesn't merge, full stop.

2. Testing Objectives

- Catch regressions before they reach `develop` or `main`, not after a teammate (or a marker) finds them.
- Verify that each use case actually does what its functional requirements say it should, not just that the code runs without throwing an error.
- Work toward the Maintainability quality requirement from the SRS, an 80% automated test coverage target, so the codebase stays safe to change as the project grows.
- Make sure critical flows (auth, listing creation, moderation, messaging) keep working even as unrelated parts of the system change underneath them.
- Keep the system accessible, WCAG 2.2 AA as the baseline, checked with real Lighthouse audits rather than just assumed.

3. Testing Types

3.1 Unit Testing

Backend unit tests use Jest and test a single service or controller in isolation, with its dependencies (repositories, other services) mocked out. These are meant to be fast and to test logic, not infrastructure, things like validation rules, price calculations, or status transitions on a listing.

Frontend unit tests also use Jest, together with React Testing Library, and focus on individual components: does a form show a validation error when it should, does a button call the right handler, that kind of thing.

> **New for Demo 4:** the three wow factors added this sprint are algorithm-heavy rather than CRUD-heavy, so unit testing carries more of the correctness burden for them than it does elsewhere in the codebase. Edge cases specifically covered: for the Bundle Optimizer, zero seller overlap (falls back to naive cheapest-per-book), full overlap (one seller covers every required book), partial overlap, and a module with no approved listings at all; for AI-Assisted Listing Creation, the OCR-to-`Book` fuzzy-matching logic (no match found, multiple close matches, garbled OCR output); and for Real-Time Textbook Auctions, bid validation (below minimum increment, seller bidding on their own auction, a bid placed after close).

3.2 Integration Testing

Backend integration tests also run through Jest, but instead of mocking the database, they run against a real, temporary Postgres container that spins up inside CI. This matters because a lot of the bugs we've actually hit (see the Coding Standards document's migration section) only show up against a real database, a mocked repository would have happily let all of them through. Integration tests cover things like a controller endpoint actually persisting a row correctly, guards correctly blocking an unauthorised role, and TypeORM relations resolving the way we expect.

> **The auction concurrency test (new for Demo 4):** Real-Time Textbook Auctions introduces the one piece of logic in this codebase that genuinely needs a concurrency guarantee, not just a correctness guarantee: two bids submitted within milliseconds of each other on the same auction must resolve deterministically, with exactly one accepted. A sequential integration test wouldn't catch a locking bug here, so the test fires two bids at the same auction simultaneously with `Promise.all([...])` against the real (ephemeral, containerised) test database, and asserts that exactly one comes back `ACCEPTED` under the `SELECT ... FOR UPDATE` row lock. This is the single test

in the suite this document would point to first if asked how a correctness property that can't be faked was actually verified.

3.3 End-to-End (E2E) Testing

Backend E2E tests use Supertest against a running instance of the full NestJS application, hitting real HTTP routes the way a client would, covering full request/response cycles including auth, guards, and database writes.

Cypress e2e testing, as of Demo 4: Still only the same two specs as Demo 3:

- **Login** – covers a full login attempt, client-side validation errors, a rejected login with wrong credentials, the password visibility toggle, and navigation to and from the login page.
- **Registration** – covers Steps 1 to 3 of the four-step flow (personal details, university email, password). Step 4 (OTP verification) isn't automated yet, since that needs the test run to read the OTP out of a Mailtrap inbox rather than a person's real email, and that integration isn't wired up.

No new Cypress specs were added this sprint, including for the three Demo 4 wow factors (UC11 AI-Assisted Listing Creation, UC12 Real-Time Textbook Auctions, UC13 Smart Multi-Book Bundle Optimizer). For the wow factors specifically, coverage comes from unit tests and backend integration/E2E tests only, no frontend E2E. This is a deliberate trade-off given the sprint's timeline (see the algorithm and concurrency testing called out above), not something being presented as complete.

Everything else (messaging, listing lifecycle, smart filter alerts, reports and bans, appeals, and the three Demo 4 wow factors) is still Jest and backend-E2E-only for now. This is tracked as a follow-up.

3.4 Manual and Exploratory Testing

Before approving a pull request, the reviewer doesn't just read the diff, they pull the branch (or check the relevant staging deployment) and actually click through the changed flow themselves. This has caught things automated tests missed, particularly UI states that only show up with specific data (empty states, long text overflow, etc.).

3.5 Accessibility Testing

Each major page is audited with Google Lighthouse against WCAG 2.2 AA as the baseline target. Current scores, taken from our own Lighthouse runs:

Page	**Accessibility Score**
Landing Page	94 / 100
Listings Browse Page	96 / 100
Create Listing Page	94 / 100
In-App Messaging	95 / 100
My Listings Page	96 / 100
Wishlist Page	96 / 100
Notifications Page	94 / 100
Saved Searches Page	96 / 100
HelpMenu Page	96 / 100
Admin Review Dashboard	95 / 100
Audit Log Page	85 / 100
Auction Page (new, UC12)	96 / 100
Bundle Optimizer Results Page (new, UC13)	90 / 100
AI-Assisted Listing (Scan/Crop) (new, UC11)	94 / 100

The two lower scorers among the audited pages, Listings Browse and Audit Log, are both pages with a lot of interactive and tabular content on screen at once (filters, grids, data tables), which naturally gives Lighthouse more contrast and labelling checks to fail on. These are tracked as follow-up items rather than being treated as resolved. The three new wow-factor pages have not been through a Lighthouse run at the time of writing; they're listed here as `Pending` rather than silently left off the table.

4. Testing Tools and Environments

Tool	Used for	Notes
Jest	Backend unit + integration tests, frontend unit tests	Runs in `ci.yml` on every push and PR
Supertest	Backend E2E tests	Hits real routes on a running NestJS instance
Postgres (containerised)	Integration/E2E test database	Ephemeral, spun up fresh inside CI, not the staging or production database
React Testing Library	Frontend component tests	Used alongside Jest
Cypress	Frontend E2E tests	Login and Registration (Steps 1–3) specs only, unchanged since Demo 3. Not extended to the Demo 4 wow factors this sprint; remaining flows are to be built.
Codecov	Coverage tracking	Separate flags for backend and frontend
SonarCloud	Static analysis, code smells, security hotspots	Runs on every push/PR to `main` and `develop`
Lighthouse	Accessibility auditing	Run manually against deployed pages
GitHub Actions	CI/CD orchestration	Runs all of the above and gates merges on the result

4.1 Environments

- **Local:** Docker Compose, run against a local Postgres container, used for day-to-day development before anything is pushed.
- **CI (ephemeral):** A fresh Postgres container created for the lifetime of the pipeline run only, used for integration and E2E tests. Nothing here ever touches staging or production data.
- **Staging:** The `develop` branch auto-deploys here. This is where manual/exploratory testing of a feature happens before it's considered safe to promote to production.

- **Production:** The `main` branch. Nothing merges here without having already passed through staging first.

5. Coverage Targets

The SRS sets an 80% automated test coverage target under the Maintainability quality requirement. We track actual coverage through Codecov, with separate badges/flags for backend and frontend so one side doing well doesn't hide the other side lagging behind.

> **Being honest about where we are:** 80% is the target we're working toward, not a hard gate that currently blocks merges on its own. Right now, merge is gated on the test suite passing and SonarCloud's Quality Gate, not on a specific coverage percentage threshold. Tightening this into an enforced minimum is a planned follow-up rather than something already in place.

6. Defect Management Process

We track bugs as GitHub Issues on our project board (`github.com/orgs/COS301-SE-2026/projects/64`) the same board we use for feature work, rather than keeping a separate bug list somewhere else.

6.1 Workflow

1. **Report:** Anyone who finds a bug (a team member, a mentor, or a marker during a demo) opens a GitHub Issue with the `bug` label, describing the steps to reproduce it and the expected vs. actual behaviour.
2. **Triage:** The issue is assigned a severity (see below) and assigned to whoever owns the relevant feature area.
3. **Fix:** A fix branch is created off `develop`, following the same `fix/short-description` naming convention as the Coding Standards document, and a regression test is written that would have caught the bug in the first place, not just a fix without a test behind it.
4. **Review and merge:** The fix goes through the normal PR process, one reviewer, `ci-success` passing, before it merges into `develop`.
5. **Verify:** The reporter (or the fix's author, if the reporter isn't available) confirms the fix actually works on staging before the issue is closed.

6.2 Severity Levels

Severity	Meaning	Expectation
Critical	System is down, or a core flow (auth, listing creation, moderation) is broken for everyone	Fixed immediately, before other work continues
High	A feature is broken or badly degraded, but there's a workaround or it's not core	Fixed within the current sprint
Medium	A feature works but behaves incorrectly in an edge case	Fixed when the owning team member has capacity

Severity	Meaning	Expectation
Low	Cosmetic issue, typo, minor styling inconsistency	Fixed opportunistically, batched with other small fixes

7. Acceptance Criteria

Before a use case is considered "done" and demo-ready, all of the following need to be true:

- Unit tests exist for the new service/controller logic, and integration or E2E tests exist for the new endpoint(s), using a real database in CI rather than a mock.
- The feature has been manually tested against the acceptance criteria written in its GitHub Issue, not just against the happy path.
- The pull request has at least one approving review from a teammate other than the author.
- The `ci-success` check is green: lint, tests, and SonarCloud's Quality Gate all pass.
- The feature has been checked on the staging deployment (not just locally) before `develop` is merged into `main`.
- Where the feature has a UI, it's been checked against the Brand Style Guide and meets WCAG 2.2 AA, no unlabelled inputs, no insufficient colour contrast, keyboard navigable.

Uni Textbook Marketplace

In collaboration **with:**





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This document is version-controlled in the GitHub repository under `/docs/Demo 4/Testing Policy.pdf`